Driver Analysis System

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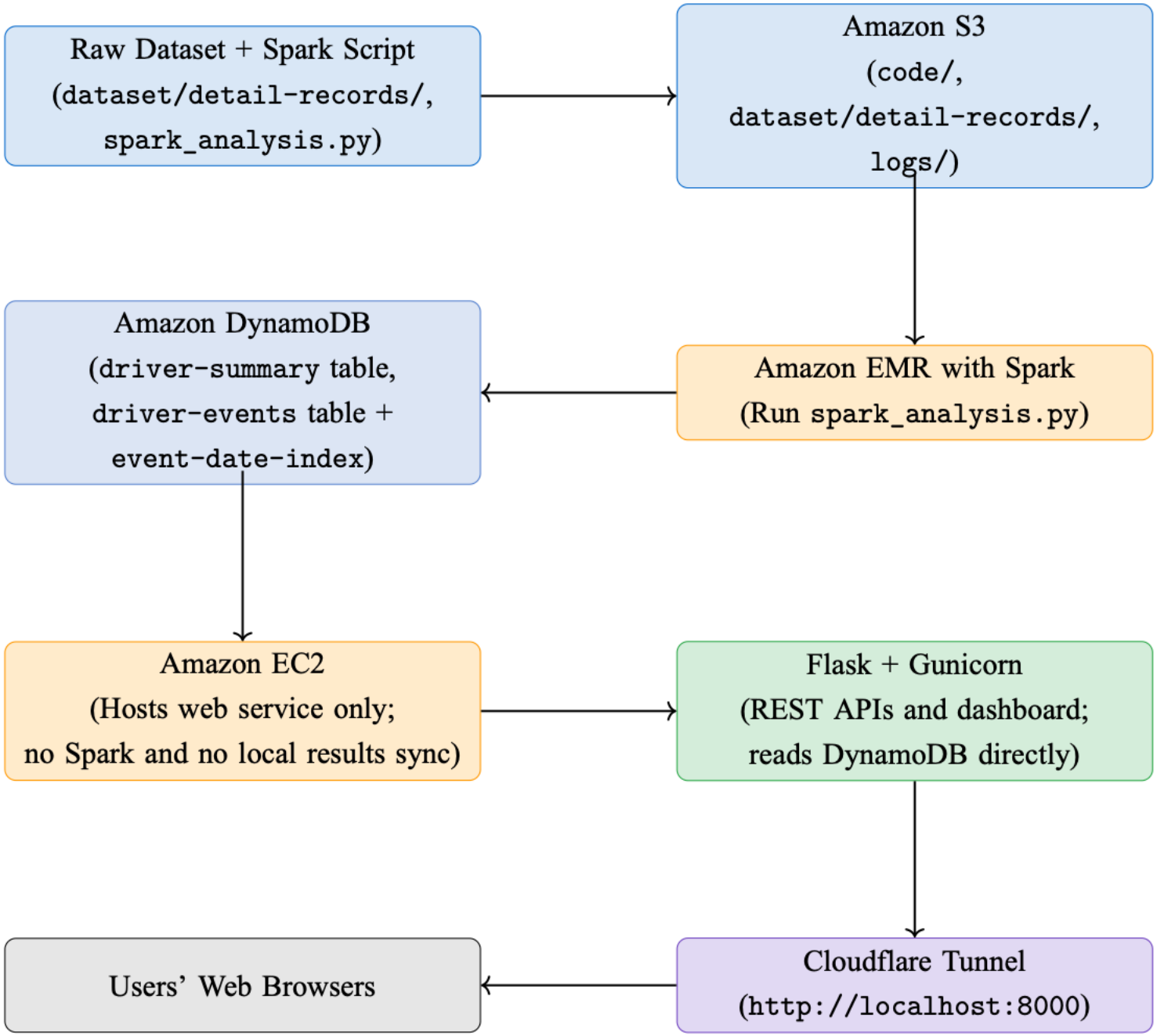
System Architecture

- Data processing module (Spark)
- Deployment backend module (AWS services)
- Backend module (Python Flask)
- Frontend module (HTML + CSS + JS)


Raw Dataset + Spark Script
(dataset/detail-records/,
spark_analysis.py)
Amazon S3
(code/,
dataset/detail-records/,
logs/)
Amazon DynamoDB
(driver-summary table,
driver-events table +
event-date-index)
Amazon EMR with Spark
(Run spark_analysis.py)
Amazon EC2
(Hosts web service only;
no Spark and no local results sync)
Flask + Gunicorn
(REST APIs and dashboard;
reads DynamoDB directly)
Users' Web Browsers
Cloudflare Tunnel
(http://localhost:8000)

Data Processing Module (Spark)

- Read `dataset/detail-records/`
- Generate driver summary
- Calculate a risk score
- Generate event records (for user-customized time period filter)
- Add DynamoDB keys
- Write into DynamoDB database

$$
\begin{aligned}
\texttt{risk_raw_score} = {} & 0.35 \times \texttt{overspeed_count} \\
& + 0.30 \times \texttt{fatigue_count} \\
& + 0.15 \times \texttt{neutral_slide_count} \\
& + 0.10 \times \texttt{rapid_speedup_count} \\
& + 0.10 \times \texttt{rapid_slowdown_count}
\end{aligned}
$$

The raw risk score is then normalized to a value between 0 and 100:

$$
\texttt{risk_score} = \frac{\texttt{risk_raw_score} - \min(\texttt{risk_raw_score})}{\max(\texttt{risk_raw_score}) - \min(\texttt{risk_raw_score})} \times 100
$$

Based on the normalized score, each driver is classified into one of three risk levels:

- **High Risk**: $\texttt{risk_score} \geq 70$
- **Medium Risk**: $\texttt{risk_score} \geq 40$ and < 70
- **Low Risk**: $\texttt{risk_score} < 40$

Backend module

- **GET** /
- **GET** /health
- **GET** /ready
- **GET** api/summary
- **GET** api/summary?**start=<>&end=<>**
- **GET** api/drivers
- **GET** api/speed/<id>
- **GET** api/speed/<id>?**limit=<>&cursor=<next>**

Frontend module

- MacOS Aqua style

- Driver behabior summary
- Per-driver speed monitor
- All drivers speed monitor

- Please refer to our demo presentation.

Deployment: **AWS Orchestration**

- Upload code and data from local to **S3**
- → run Spark jobs on **EMR**
- → store processed results in **DynamoDB**
- → run Flask/Gunicorn on **EC2**
- → expose to the public internet via **Cloudflare Tunnel**

Deployment: **AWS Orchestration**

- **S3:** Stores Spark scripts, raw datasets, and EMR logs.
- **EMR:** Runs `spark/spark_analysis.py` for data processing and analysis.
- **DynamoDB:** Stores processed summaries and event-level details generated by Spark.
- **EC2:** Runs only the Flask web application (no Spark jobs).
- **Gunicorn:** Serves as the production WSGI server for Flask.
- **Cloudflare Tunnel: Exposes** the EC2 local service (`localhost:8000`) to a public domain.

Deployment: **Security Group**

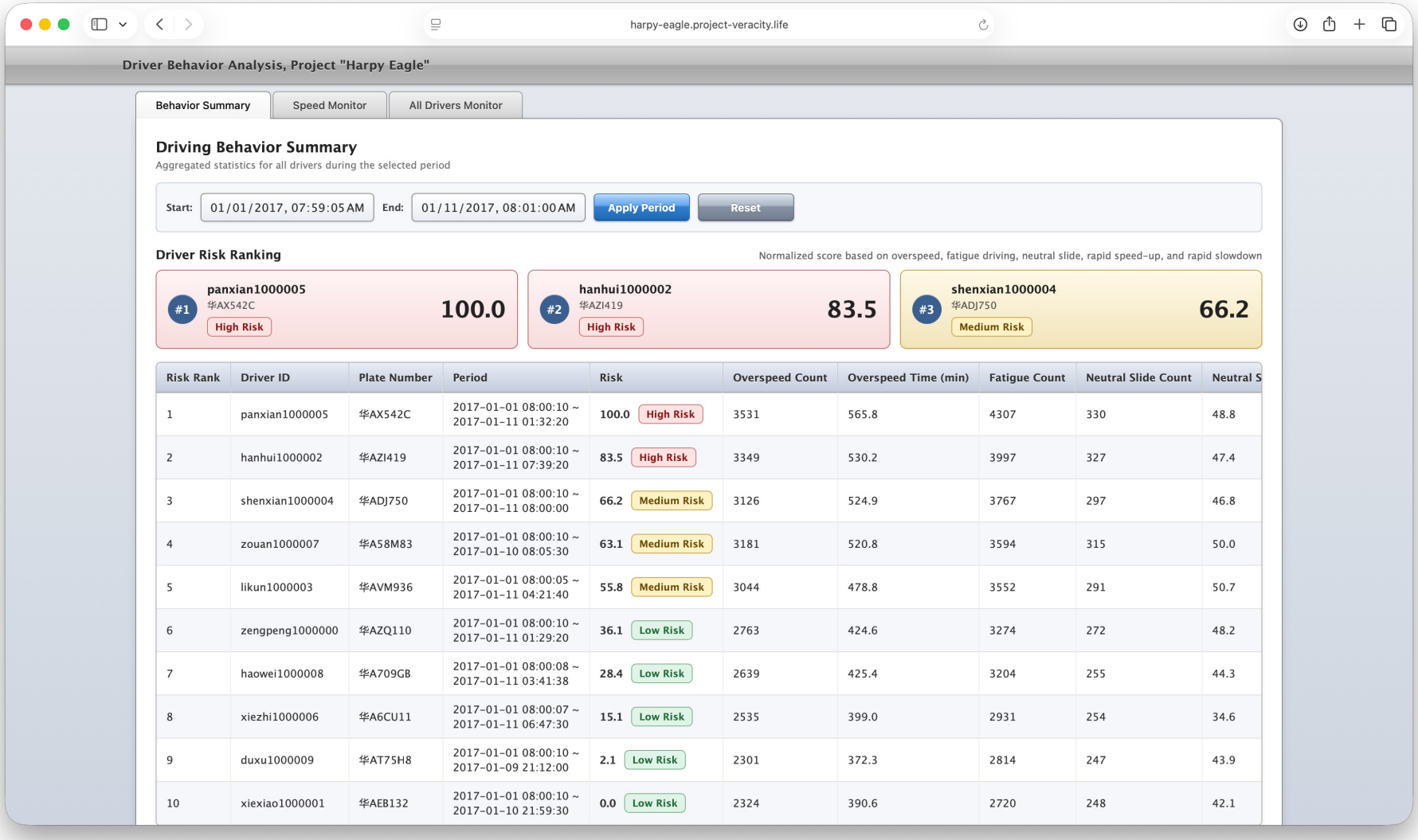
- The **security group** only needs to allow SSH access for **administrators**, since public traffic goes through Cloudflare Tunnel and there is no need to open ports `80/443` directly.

Deployment: **IAM Configuration**

- IAM: EMR instance profile = S3 read/log write + DynamoDB read/write; EC2 role = DynamoDB read only

Demo

https://harpy-eagle.project-veracity.life



Risk Rank	Driver ID	Plate Number	Period	Risk	Overspeed Count	Overspeed Time (min)	Fatigue Count	Neutral Slide Count	Neutral S
1	panxian1000005	华AX542C	2017-01-01 08:00:10 ~ 2017-01-11 01:32:20	100.0 High Risk	3531	565.8	4307	330	48.8
2	hanhui1000002	华AZI419	2017-01-01 08:00:10 ~ 2017-01-11 07:39:20	83.5 High Risk	3349	530.2	3997	327	47.4
3	shenxian1000004	华ADJ750	2017-01-01 08:00:10 ~ 2017-01-11 08:00:00	66.2 Medium Risk	3126	524.9	3767	297	46.8
4	zouan1000007	华A58M83	2017-01-01 08:00:10 ~ 2017-01-10 08:05:30	63.1 Medium Risk	3181	520.8	3594	315	50.0
5	likun1000003	华AVM936	2017-01-01 08:00:05 ~ 2017-01-11 04:21:40	55.8 Medium Risk	3044	478.8	3552	291	50.7
6	zengpeng1000000	华AZQ110	2017-01-01 08:00:10 ~ 2017-01-11 01:29:20	36.1 Low Risk	2763	424.6	3274	272	48.2
7	haowei1000008	华A709GB	2017-01-01 08:00:08 ~ 2017-01-11 03:41:38	28.4 Low Risk	2639	425.4	3204	255	44.3
8	xiezhi1000006	华A6CU11	2017-01-01 08:00:07 ~ 2017-01-11 06:47:30	15.1 Low Risk	2535	399.0	2931	254	34.6
9	duxu1000009	华AT75H8	2017-01-01 08:00:10 ~ 2017-01-09 21:12:00	2.1 Low Risk	2301	372.3	2814	247	43.9
10	xiexiao1000001	华AEB132	2017-01-01 08:00:10 ~ 2017-01-10 21:59:30	0.0 Low Risk	2324	390.6	2720	248	42.1